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1.12 (a) The data were obtained observational.

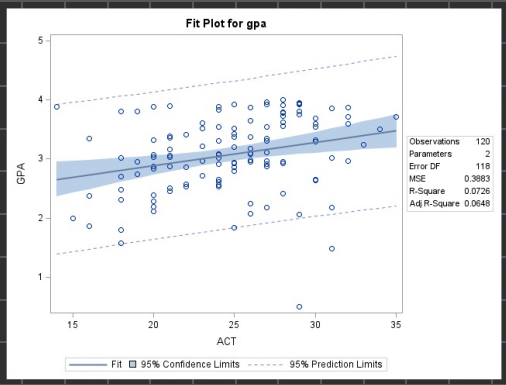
(b) I do not think the conclusions were valid, since there could be other reasons causing the same effect as stated in (c).

(c) By ChatGPT, dietary habits, stress levels, vaccination status could be some variables affecting both factors.

(d) Construct two groups randomly, one of them exercises a given hour while we do not change the other groups' exercise habit.

1.19 (a) $\hat{\beta}_0 = 2.11405$, $\hat{\beta}_1 = 0.03883 \Rightarrow E(Y) = 2.11405 + 0.03883X$
(Parameter Estimates Table)

(b) Although the scatter plot looks like it doesn't have a pattern, the F-value is large in terms of $H_0: \beta_1 = 0$, $H_1: \beta_1 \neq 0$, so it fits well.



(c) $\hat{Y} = 2.11405 + 0.03883 \times 30 = 3.27895$

(d) The change in the mean response when the entrance test score increase by one point is $\hat{\beta}_1 = 0.03883$.

2.13 (a) $s_{\hat{Y}}^2 = MSE\left(\frac{1}{n} + \frac{(x_i - \bar{x})^2}{n \cdot \sum(x_i^2) - n\bar{x}^2}\right) = 0.38828\left(\frac{1}{120} + \frac{(28 - 24.725)^2}{120 \cdot 631.158 - 120 \cdot 24.725^2}\right) \doteq 0.0049$

$t_{0.025}(118) \doteq 1.9803$, $t_{0.025}(118) s_{\hat{Y}} \doteq 0.1398$

$\Rightarrow$ mean response 95% C.I. of test score 28

$\Rightarrow [3.2012 \pm 0.1398] = [3.0613, 3.3410]$

(b) $S_{pred}^2 = MSE\left(1+\frac{1}{n}+\frac{(x-\bar{x})^2}{S_{xx}}\right) \doteqdot 0.3932$

因此, 95% C.I. of test score 28 $\Rightarrow [2.3099,\ 3.5844]$

(c) Yes, (b) is wider than (a), since the error of mean is smaller, C.I. is more narrow

2.23 (a)

	DoF	SS	MS	F-value
Regression	1	3.58785	3.58785	9.24
Error	118	45.81761	0.38828	
Total	119	49.40545		

(b) The alternative hypothesis is $\beta_1 \neq 0$ (while $H_0: \beta_1 = 0$),

decision rule : $\{ F > F_{0.01}(1, 118) \}$, and yes, we reject H_0.

Codes : github.com/doggydoggy0101/math_project